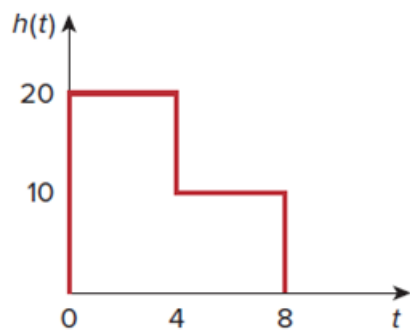


## Problem

Find the Laplace transform of the function  $h(t)$  in Fig. 15.6.

**Figure 15.6:**



## Step-by-step solution

**Step 1 of 3**

Consider the signal  $h(t)$  shown in Figure 1.

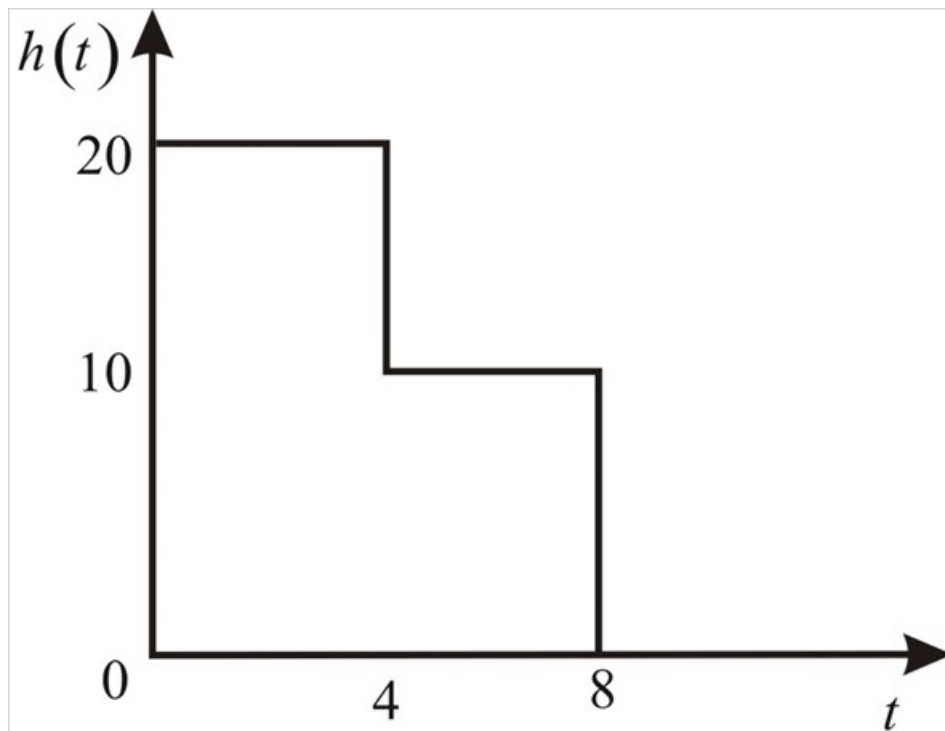


Figure 1

**Step 2 of 3**

The signal  $h(t)$  is mathematically represented as,

$$\begin{aligned} h(t) &= 20[u(t) - u(t-4)] + 10[u(t-4) - u(t-8)] \\ &= 20u(t) - 10u(t-4) - 10u(t-8) \end{aligned}$$

The Laplace transform of the signal  $h(t)$  is,

$$\mathcal{L}(h(t)) = \mathcal{L}[20u(t) - 10u(t-4) - 10u(t-8)]$$

Linearity property of Laplace transform is given by,

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = \mathcal{L}[a_1 f_1(t)] + \mathcal{L}[a_2 f_2(t)]$$

Use Linearity property to determine the Laplace transform of the signal  $h(t)$ .

$$\mathcal{L}[20u(t) - 10u(t-4) - 10u(t-8)] = \begin{bmatrix} \mathcal{L}[20u(t)] - \mathcal{L}[10u(t-4)] \\ -\mathcal{L}[10u(t-8)] \end{bmatrix} \dots\dots (1)$$

Here, consider  $f(t) = u(t)$ , then its Laplace transform is  $\mathcal{L}(f(t)) = F(s)$ ,

$$\begin{aligned} \mathcal{L}(f(t)) &= F(s) \\ &= \int_0^{\infty} u(t) e^{-st} dt \\ &= \int_0^{\infty} e^{-st} dt \\ &= \left. \frac{e^{-st}}{-s} \right|_0^{\infty} \\ &= -\frac{1}{s} (e^{-s\infty} - e^{-s \cdot 0}) \\ &= -\frac{1}{s} (0 - 1) \\ &= \frac{1}{s} \end{aligned}$$

### Step 3 of 3

Time shifting property of Laplace transform is given by,

$$\mathcal{L}[(f(t-a))u(t-a)] = e^{-as} F(s)$$

After applying shifting property to Laplace equation (1), here substitute  $F(s) = \frac{1}{s}$ ,

$$\begin{aligned} \mathcal{L}[20u(t) - 10u(t-4) - 10u(t-8)] &= 20F(s) - 10e^{-4s}F(s) - 10e^{-8s}F(s) \\ &= 20\frac{1}{s} - 10e^{-4s}\frac{1}{s} - 10e^{-8s}\frac{1}{s} \\ &= \frac{10}{s}(2 - e^{-4s} - e^{-8s}) \end{aligned}$$

Therefore, the Laplace transform of the signal  $h(t)$  is  $\boxed{\frac{10}{s}(2 - e^{-4s} - e^{-8s})}$ .